

AEGIS: A Byzantine-Resilient Federated Learning Protocol for Highly Non-IID Environments

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Abstract—Federated Learning (FL) is vulnerable to Byzantine attacks, where compromised clients submit poisoned updates to degrade the global model. Existing robust aggregators often struggle to distinguish between adversarial behavior and the natural statistical drift inherent to highly Non-IID datasets. We introduce *Aegis*, an aggregation protocol designed for environments with high data skew. Operating with an asymptotic complexity of $\mathcal{O}(kd)$, *Aegis* replaces standard distance-based filters with a two-pass dual-metric anomaly detection system, utilizing adaptive Median Absolute Deviation (MAD) thresholding and cross-round reputation tracking. We evaluate *Aegis* on CIFAR-10 under a 30% Byzantine threat model. Results show *Aegis* mitigates conventional attacks (Sign-Flipping, Label-Flipping, Volume Spam), maintaining global accuracy within 5% of the baseline. We also evaluate the computational scaling and limits of statistical filtering: while the two-pass architecture improves resilience against Inner Product Manipulation (IPM), variance-envelope attacks (such as ALIE) bypass per-round filters entirely, and the constant factors of multi-pass median decontamination yield higher wall-clock execution times at small client scales compared to traditional pairwise methods.

Index Terms—Federated Learning, Byzantine Resilience, Model Poisoning, Non-IID Data, Robust Aggregation.

I. INTRODUCTION

Federated Learning (FL) enables edge devices to collaboratively train a shared global model without exchanging raw, privacy-sensitive data. However, this decentralized architecture introduces a critical attack vector: Byzantine clients. Whether compromised by attackers or suffering from hardware faults, these nodes can transmit poisoned model updates to degrade accuracy, insert backdoors, or prevent convergence [2].

Securing FL is complicated by the statistical heterogeneity of real-world edge data. In highly Non-IID environments, legitimate clients often produce weight updates that diverge from the global average simply due to local data skew. Traditional Byzantine-robust aggregators frequently penalize this statistical diversity, inadvertently discarding valuable updates from legitimate nodes. Furthermore, distance-based methods like Krum inherently scale quadratically $\mathcal{O}(k^2d)$ with the number of participating clients, posing theoretical bottlenecks for massively distributed swarms.

We propose **Aegis** to address these limitations. The protocol separates adversarial anomalies from natural Non-IID variance using a two-pass median decontamination process,

historical reputation tracking, and adaptive credit scoring. Rather than relying on a single spatial metric, *Aegis* evaluates Euclidean distance and Cosine similarity simultaneously. It applies MAD-based adaptive thresholds to these metrics, automatically adjusting to the swarm’s variance each round. In our evaluation, we demonstrate robust defense against conventional attacks, compare empirical execution dynamics against baseline aggregators, and explicitly document the limits of statistical filtering against advanced optimization attacks like ALIE and IPM.

II. RELATED WORK

The pursuit of Byzantine resilience in FL has yielded several families of defense mechanisms, each with specific tradeoffs.

Distance-Based Aggregators: Krum and Multi-Krum [2] select updates using pairwise distances. They provide provable resilience but theoretically require $\mathcal{O}(k^2d)$ computations. Crucially, distance selection frequently fails against optimization-based attacks like ALIE [5]. Bulyan [9] attempts to fix Krum’s vulnerabilities via a two-phase meta-aggregation, but the $\mathcal{O}(k^3d)$ cost is prohibitive.

Statistical and Spectral Methods: Trimmed Mean and Coordinate-wise Median [3] operate efficiently but assume coordinate-wise independence, leaving them vulnerable to Inner Product Manipulation (IPM) attacks [6]. Divide-and-Conquer (DnC) [12] employs spectral methods to detect correlated perturbations but incurs significant computational overhead. A recent meta-analysis [13] confirms that optimization-based attacks continue to bypass standard statistical rules.

Trust and Hybrid Defenses: FLTrust [4] and Zeno [10] score updates against a server-side root dataset. This approach violates pure FL privacy norms and struggles in extreme Non-IID settings where the server data may not reflect edge distributions. FLAME [8] combines cosine-based clustering and adaptive clipping, serving as a precedent for angular pre-screening, but relies on $\mathcal{O}(k^2d)$ HDBSCAN clustering. *Aegis* targets resilience through robust filtering and reputation tracking without relying on global momentum buffers or external clean datasets.

III. THREAT MODEL AND ENVIRONMENT

A. System Architecture and Data Heterogeneity

We consider a cross-device FL architecture with a central parameter server and $n = 30$ clients. To simulate realistic data skew, we use Shard-based partitioning [1]. The dataset is sorted by label, and each client is allocated exactly 4 random data shards, ensuring no single client holds a representative sample of the global distribution.

B. Adversary and Participation Model

We evaluate Aegis under a *Fixed Adversary with Random Participation* model. The identities of compromised nodes are fixed at the start of training. During each round, a random subset of clients $m \in [15, 25]$ is selected. Consequently, the exact number of participating Byzantine clients follows a Hypergeometric distribution:

$$X \sim \text{Hypergeometric}(N = 30, K = 9, m) \quad (1)$$

where $K = 9$ represents a 30% global Byzantine presence.

C. Attack Hierarchy

We evaluate the protocol against a spectrum of attacks:

- 1) **Untargeted Sign-Flipping & Additive Noise:** Malicious clients negate gradients or inject Gaussian noise.
- 2) **A Little Is Enough (ALIE) [5]:** Gradients are crafted strictly within the variance envelope of benign updates.
- 3) **Inner Product Manipulation (IPM) [6]:** Attackers craft vectors forcing the global update into gradient ascent.
- 4) **Targeted Label-Flipping & Volume Spam:** Adversaries deterministically flip labels and artificially inflate their reported dataset size n_k .

IV. THE AEGIS PROTOCOL: MECHANICS & ARCHITECTURE

Aegis executes a 6-step pipeline per round (summarized in Algorithm 1 and Figure 1). By utilizing Introspect for median computations and avoiding pairwise distance matrices, Aegis maintains an asymptotic time complexity of $\mathcal{O}(kd)$.

A. Step 1: Distribution & Local Training

The server broadcasts W_{global}^t . Client k trains locally and returns its updated weights W_k and claimed sample size n_k . We compute the local pseudo-gradient: $\Delta_k = W_k - W_{\text{global}}^t$.

B. Step 2: Two-Pass Robust Center

Under optimization attacks like IPM ($\epsilon = 1.0$), Byzantine gradients bias the standard coordinate-wise median. This creates a contaminated reference point, leading to false-positive rejections of honest non-IID clients. Aegis addresses this via a two-pass decontamination process.

- 1) **Pass 1 (Pre-screen):** Compute a preliminary median $\Delta_{\text{pre}} = \text{median}_{\text{all}}(\Delta_k)$. Compute cosine similarities against Δ_{pre} and provisionally remove clients where $\cos < 0$.
- 2) **Pass 2 (Clean Anchor):** Recompute the final robust center solely from the surviving clients: $\Delta_{\text{median}} = \text{median}_{\text{clean}}(\Delta_k)$.

Algorithm 1 AEGIS: Two-Pass Byzantine-Resilient Aggregation with Soft Directional Penalty

Require: Updates $\{(W_k, n_k)\}_{k=1}^K$, global model W_{global}^t , round t , reputation state $\{R_k\}$
Ensure: Aggregated model W_{global}^{t+1}

- 1: **// Step 1: Compute Pseudo-Gradients**
- 2: **for** each client k **do**
- 3: $\Delta_k \leftarrow W_k - W_{\text{global}}^t$
- 4: **end for**
- 5:
- 6: **// Step 2: Two-Pass Robust Center (Median Decontamination)**
- 7: $\Delta_{\text{pre}} \leftarrow \text{median}_k(\Delta_k)$ $\triangleright$ Pass 1: Preliminary median
- 8: **for** each client k **do**
- 9: $\cos_k^{\text{pre}} \leftarrow \frac{\Delta_k \cdot \Delta_{\text{pre}}}{\|\Delta_k\|_2 \|\Delta_{\text{pre}}\|_2}$
- 10: **end for**
- 11: $\mathcal{S} \leftarrow \{k : \cos_k^{\text{pre}} \geq \tau_{\text{pass1}}\}$ $\triangleright$ Screen adversarial directions
- 12: $\Delta_{\text{med}} \leftarrow \text{median}_{k \in \mathcal{S}}(\Delta_k)$ $\triangleright$ Pass 2: Debiased clean median
- 13:
- 14: **// Step 3: Dual Anomaly Scoring (all clients re-scored)**
- 15: **for** each client k **do**
- 16: $E_k \leftarrow \|\Delta_k - \Delta_{\text{med}}\|_2$ $\triangleright$ Euclidean distance
- 17: $\text{Cos}_k \leftarrow \frac{\Delta_k \cdot \Delta_{\text{med}}}{\|\Delta_k\|_2 \|\Delta_{\text{med}}\|_2}$ $\triangleright$ Cosine similarity
- 18: $P_k \leftarrow 1.0 - \text{Cos}_k$ $\triangleright$ Cosine penalty (continuous)
- 19: **end for**
- 20:
- 21: **// Step 4: Adaptive MAD Gatekeeper (Euclidean only)**
- 22: $E_{\text{med}} \leftarrow \text{median}(E_k)$; $D \leftarrow \text{median}(|E_k - E_{\text{med}}|)$
- 23: $K_{\text{phase}} \leftarrow K_{\text{max}} - (K_{\text{max}} - K_{\text{min}}) \cdot \min\left(\frac{t}{t_{\text{warmup}}}, 1\right)$
- 24: $CV \leftarrow D / (E_{\text{med}} + \epsilon)$
- 25: $K \leftarrow \max(K_{\text{floor}}, K_{\text{phase}} \cdot (1 + \eta \cdot CV))$
- 26: $T \leftarrow E_{\text{med}} + K \cdot D$
- 27: $\mathcal{A} \leftarrow \{k : E_k \leq T\}$ $\triangleright$ Magnitude gate only; no hard cosine gate
- 28:
- 29: **// Step 5: Volume Bounding, Reputation & Soft Credit Scoring**
- 30: **for** each $k \in \mathcal{A}$ **do**
- 31: $v_k \leftarrow \min(n_k, 2.0 \times \text{median}_{j \in \mathcal{A}}(n_j))$ $\triangleright$ Volume clip
- 32: $R_k \leftarrow \gamma \cdot R_k + (1 - \gamma) \cdot P_k$ $\triangleright$ Reputation update (EMA)
- 33: $C_k \leftarrow \frac{v_k}{E_k + \alpha \cdot P_k + \lambda \cdot R_k + \epsilon}$ $\triangleright$ Soft directional suppression
- 34: **end for**
- 35: $\beta_k \leftarrow C_k / \sum_{j \in \mathcal{A}} C_j$ $\triangleright$ Normalized weights
- 36:
- 37: **// Step 6: Aggregation**
- 38: $W_{\text{global}}^{t+1} \leftarrow \sum_{k \in \mathcal{A}} \beta_k \cdot W_k$
- 39: **return** W_{global}^{t+1}

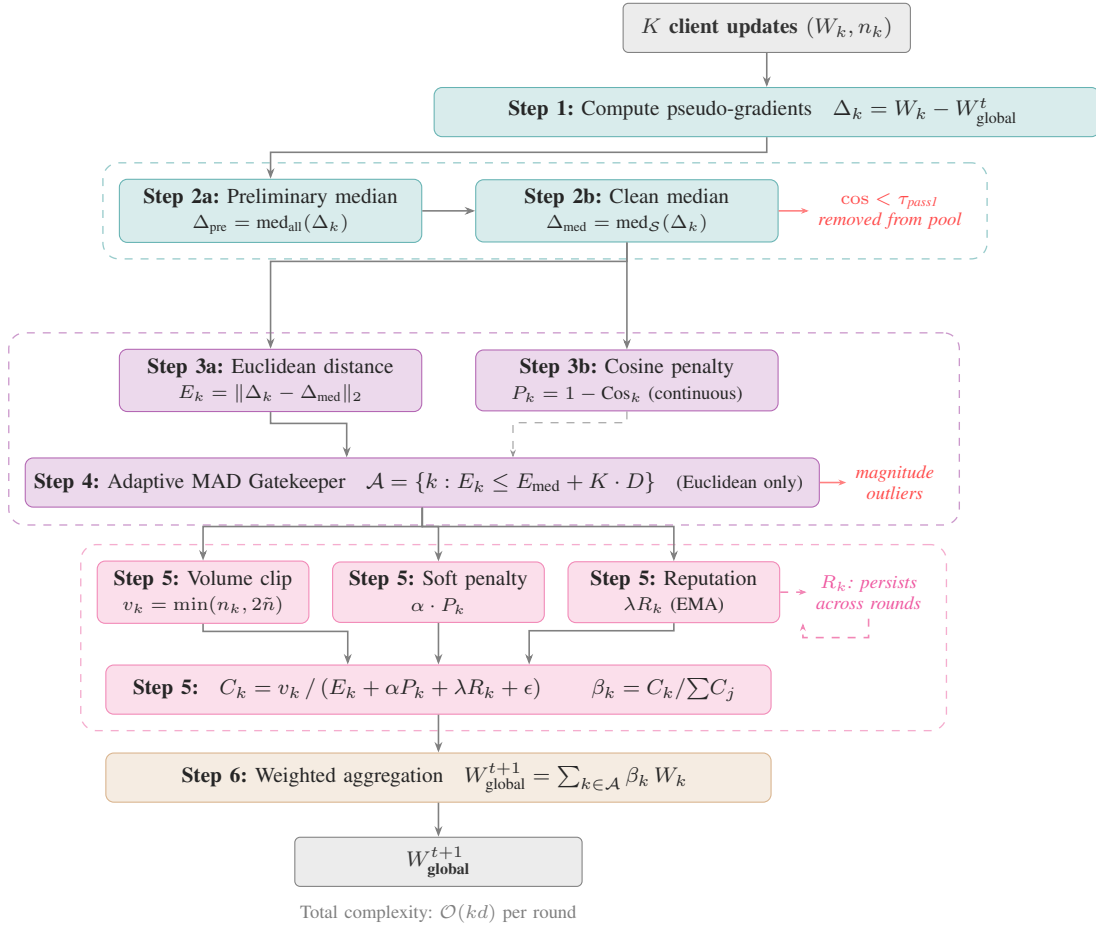


Figure 1. AEGIS aggregation pipeline. Step 2 decontaminates the coordinate-wise median via two-pass cosine pre-screening ($\tau_{\text{pass1}} = -0.3$). Step 3 calculates dual-metric anomaly scores. Step 4 applies an adaptive MAD-based Euclidean filter as the sole hard gate. The cosine penalty P_k bypasses the hard gate (dashed arrow) and enters the soft credit score in Step 5, where it combines with volume bounds and cross-round reputation R_k to produce normalized aggregation weights β_k . No hard directional threshold is applied; directional suppression is entirely continuous via αP_k in the credit score denominator.

C. Step 3: Dual Anomaly Scoring

Every client is evaluated against Δ_{median} . First, Euclidean Distance measures magnitude to detect Additive Noise:

$$E_k = \|\Delta_k - \Delta_{\text{median}}\|_2 \quad (2)$$

Second, Cosine Similarity measures direction to detect Sign and Label-Flipping:

$$\text{Cos}_k = \frac{\Delta_k \cdot \Delta_{\text{median}}}{\|\Delta_k\|_2 \|\Delta_{\text{median}}\|_2} \quad \text{and} \quad P_k = 1.0 - \text{Cos}_k \quad (3)$$

D. Step 4: Adaptive Thresholding & Gatekeeper Logic

Aegis self-calibrates its rejection threshold using Median Absolute Deviation (MAD). Base metrics are: $E_{\text{median}} = \text{median}(E_k)$ and $D = \text{median}(|E_k - E_{\text{median}}|)$. An adaptive multiplier (K) decays over a warmup period, modulated by the swarm variance ($CV = D/E_{\text{median}}$):

$$K_{\text{phase}} = K_{\text{max}} - (K_{\text{max}} - K_{\text{min}}) \times \left(\frac{t}{\text{warmup_rounds}} \right) \quad (4)$$

$$K = \max(K_{\text{safe_floor}}, K_{\text{phase}} \times (1 + \text{Var_Sensitivity} \times CV)) \quad (5)$$

The rejection threshold is $T = E_{\text{median}} + K \cdot D$. Updates are discarded solely based on magnitude if $E_k > T$. Unlike earlier iterations of this protocol, the hard directional gate has been removed to prevent the collateral exclusion of honest but highly skewed clients.

E. Step 5: Cross-Round Reputation Tracking & Credit Scoring

To penalize persistent stealth attackers, the server maintains an exponential moving average (EMA) reputation penalty R_k for each client.

$$R_k^{t+1} = \gamma R_k^t + (1 - \gamma) P_k^t \quad (6)$$

To prevent Volume Spam, claimed sizes are clipped relative to the median of approved clients. This prevents a single spammer from skewing a mean-based bound:

$$v_k = \min(n_k, 2.0 \times \text{median}(n_{\text{approved}})) \quad (7)$$

The final credit score integrates the magnitude, soft directional penalty, and historical reputation (scaled by λ):

$$C_k = \frac{v_k}{E_k + \alpha \cdot P_k + \lambda \cdot R_k + \epsilon} \quad (8)$$

Architectural Note: Soft Directional Penalty. Earlier versions of Aegis employed a hard directional gate ($\text{Cos}_k \geq 0$) alongside the Euclidean filter. Under pathological Non-IID conditions (4 shards/client), this binary threshold inadvertently rejected 20–30% of honest clients whose cosine similarity with the median naturally fluctuated near zero due to class specialization. We replace this hard gate with a continuous soft penalty: the cosine penalty P_k enters the credit score denominator scaled by $\alpha = 30$. A sign-flip attacker ($\text{Cos}_k \approx -1$, $P_k \approx 2.0$) receives a penalty term of $\alpha \cdot P_k = 60$, reducing its normalized weight to $\sim 7\%$ of the aggregation. Meanwhile, an honest Non-IID client with slight negative alignment ($\text{Cos}_k = -0.05$, $P_k = 1.05$) receives a $5\times$ weight reduction compared to a well-aligned client, avoiding total exclusion. This eliminates step-function false positives while maintaining equivalent attacker suppression.

Normalized weights are $\beta_k^t = C_k / \sum C_k$.

F. Step 6: Global Aggregation

The final step applies the normalized credit scores to the approved client updates to form the new global model:

$$W_{global}^{t+1} = \sum \beta_k^t W_k \quad (9)$$

By explicitly filtering out adversaries and weighting clients based on geometrical trust, the weighted average provides stable convergence.

G. Computational Complexity Analysis

Each step of AEGIS operates in $\mathcal{O}(kd)$ time, where k is the number of participating clients and d is the model dimension. The two-pass median computation (Step 2) uses Introselct—an introspective hybrid of Quickselect and Median-of-Medians—guaranteeing $\mathcal{O}(k)$ time per coordinate across d coordinates. The dual anomaly scoring (Step 3) requires one dot product and one norm computation per client, both $\mathcal{O}(d)$. The adaptive MAD thresholding (Step 4) and credit scoring (Step 5) operate on k scalar values in $\mathcal{O}(k)$ time. No step requires pairwise client comparisons, avoiding the $\mathcal{O}(k^2d)$ bottleneck of distance-based methods (Krum). The total per-round asymptotic complexity is $\mathcal{O}(kd)$, identical to standard FedAvg.

V. EXPERIMENTAL SETUP

Model & Data: We evaluate a CNN architecture featuring GroupNorm (groups=2) and Dropout (0.25), comprising $\sim 290\text{K}$ parameters, trained on CIFAR-10. GroupNorm is used to avoid the instability of BatchNorm running statistics across Non-IID clients.

Hyperparameters: Local SGD utilizes momentum (0.8), LR (0.01), and Batch Size (32). Aegis hyperparameters are fully detailed in Table I. Evaluation uses 500 communication rounds with $n = 30$ total clients and 15-28 active clients per round.

Table I
AEGIS HYPERPARAMETERS

Symbol	Parameter	Value
<i>Two-Pass Median Decontamination</i>		
τ_{pass1}	Pass 1 cosine threshold	−0.3
<i>Adaptive MAD Thresholding</i>		
K_{max}	Initial threshold multiplier	6.0
K_{min}	Final threshold multiplier	2.0
K_{floor}	Minimum K clamp (Non-IID)	4.0
t_{warmup}	Warmup decay period	300 rounds
η	Variance sensitivity	3.0
<i>Soft Directional Penalty & Reputation</i>		
α	Cosine penalty weight	30.0
γ	Reputation EMA decay	0.95
λ	Reputation scaling factor	20.0
<i>Server Aggregation</i>		
ϵ	Numerical stability constant	10^{-9}

VI. RESULTS AND DISCUSSION

A. Baseline Convergence and Aggregator Comparison

To establish the statistical stability of the protocol in the absence of Byzantine adversaries, we evaluated AEGIS across multiple random seeds and compared its convergence trajectory to standard baseline aggregators (FedAvg, CWMed, Krum, FoolsGold, and Bulyan).

As illustrated in Fig. 2, the protocol demonstrates highly consistent convergence despite the severe data heterogeneity (4 shards per client), yielding a mean baseline accuracy of 75.53% with a standard deviation of strictly 0.18% across three seeds.

Furthermore, as shown in Fig. 3 and Fig. 4, our empirical evaluation reveals that traditional distance-based and statistical aggregators perform significantly better on clean, highly Non-IID data than previously assumed in related literature. FedAvg achieved 76.16%, AEGIS achieved 76.08%, CWMed reached 72.31%, and Krum secured 71.30%. This establishes a tight, highly competitive baseline ceiling of $\sim 71\text{--}76\%$ before the introduction of Byzantine attackers.

To complete the baseline landscape, we also evaluated FoolsGold and Bulyan under the same clean 4-shard conditions (Fig. 5 and Fig. 6). FoolsGold, designed to penalize Sybil attacks via historical cosine similarities, inherently misclassifies the natural divergence of Non-IID clients as adversarial, suppressing legitimate updates and converging to a lower clean accuracy (73.80%). Bulyan, a meta-aggregator built on Krum, inherits its spatial limitations and incurs further precision loss during its trimming phase, yielding a baseline of just 67.69%.

B. Resilience Across Conventional Attacks

Aegis was evaluated against Sign-Flipping, Targeted Label-Flipping, and Inner Product Manipulation (IPM) attacks. As shown in Fig. 7 and Fig. 8, the protocol successfully neutralizes naive poisoning and preserves global model integrity despite the high Non-IID data skew.

Notably, under a 30% Label-Flip attack, AEGIS achieves 75.13% accuracy, marginally outperforming the clean baseline

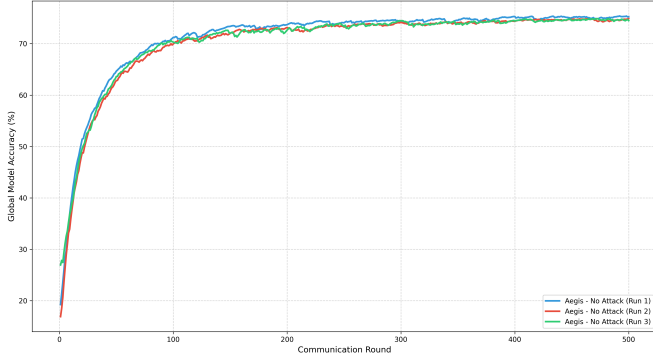


Figure 2. Global model test accuracy over 500 rounds for the clean baseline (0% Byzantine) across three independent random seeds.

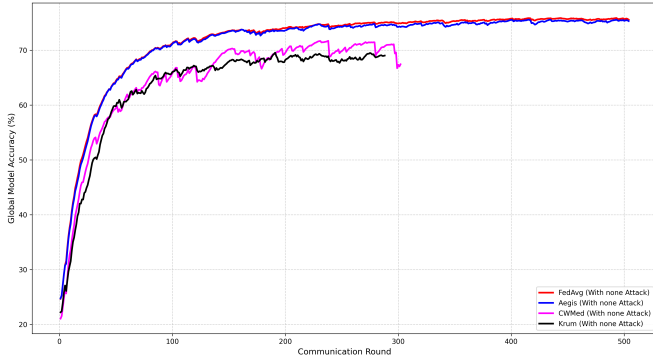


Figure 3. Test accuracy convergence of standard aggregators vs AEGIS without Byzantine attacks in a 4-shard Non-IID environment. Note that CWMed and Krum trigger early stopping prior to 500 rounds due to instability plateaus.

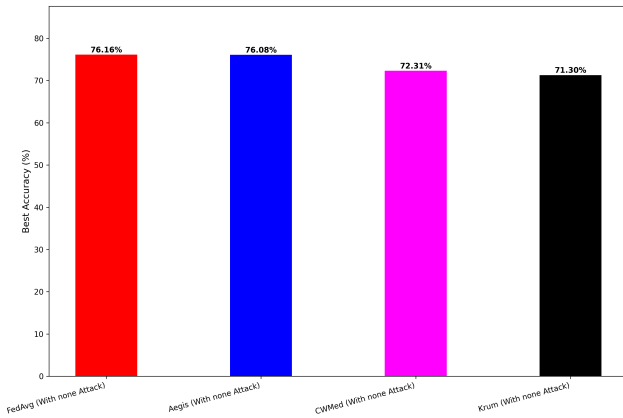


Figure 4. Final baseline accuracies without attacks. Krum and CWMed successfully surpass 71% despite the extreme data skew.

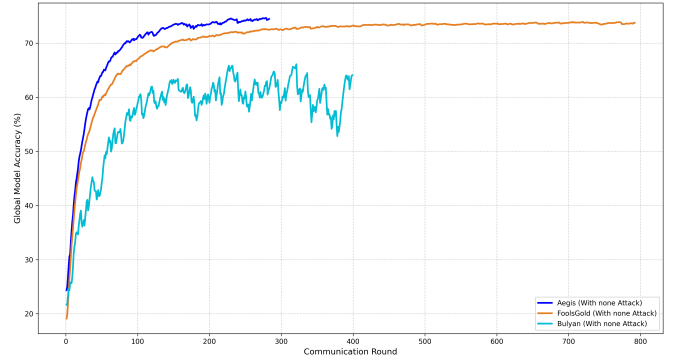


Figure 5. Baseline test accuracy comparison including FoolsGold and Bulyan without Byzantine attacks. FoolsGold penalizes natural Non-IID variance.

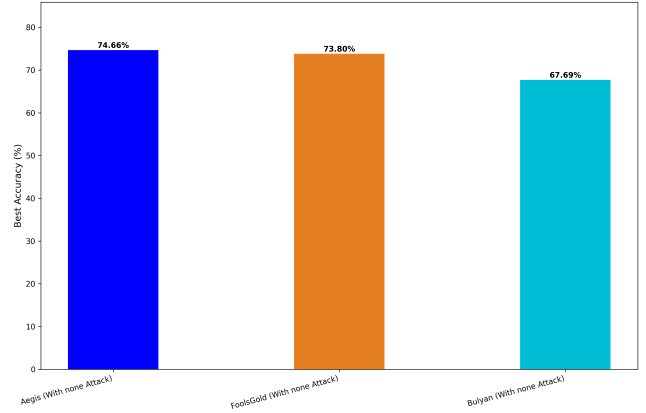


Figure 6. Final clean baseline accuracies comparing AEGIS against Sybil-centric (FoolsGold) and meta-aggregation (Bulyan) defenses.

run. This anomaly highlights the efficacy of the dual-metric screening: the targeted adversaries are filtered out entirely, leaving the remaining 70% of honest clients to stochastically converge to a slightly stronger local optimum due to the natural variance of their dataset shards. Conversely, the optimization-based IPM attack drags the model down to 64.12%, demonstrating that while Aegis limits the damage via two-pass median decontamination, mathematically scaled adversaries remain inherently harder to isolate than naive flips.

C. Advanced Threats: Volume Spam and Variance Envelopes

To evaluate the protocol's secondary defenses, we introduced a Volume Spam attack (where adversaries artificially inflate their reported dataset sizes to hijack weighted aggregation) and the A Little Is Enough (ALIE) attack.

As shown in Fig. 9, AEGIS perfectly nullifies the Volume Spam vector, achieving 74.09% accuracy. The median-based volume clipping defined in Step 5 ensures that adversaries cannot scale their updates mathematically without bounds. However, as hypothesized, the omniscient ALIE attacker successfully constructs poisoned vectors that lie entirely within the variance envelope of the honest clients' updates. By passing both the cosine pre-screen and the adaptive Euclidean



Figure 7. Test accuracy demonstrating Aegis’s resilience across conventional and optimization-based attacks. The protocol tightly bounds the impact of both Sign-Flip and IPM.

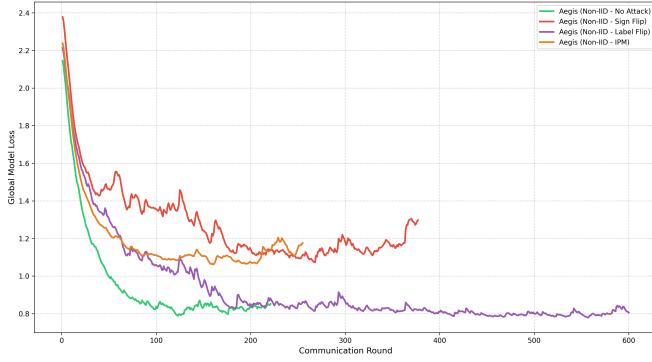


Figure 8. Test loss across communication rounds under multi-attack scenarios.

gatekeeper (registering a 0.0% detection rate in our empirical logs), ALIE bypasses the per-round filtering logic and forces the global model accuracy down to 17.39%.

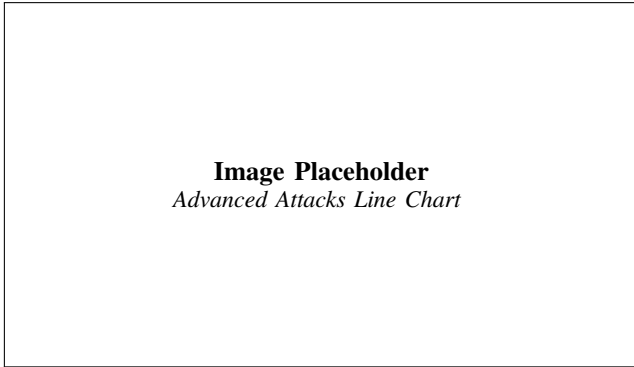


Figure 9. Performance under Volume Spam and ALIE attacks. AEGIS neutralizes volume-based hijacking but remains vulnerable to ALIE’s variance-bound optimization.

D. Comparative Evaluation

As shown in Table II, Aegis achieves 67.38% accuracy under a 30% Sign-Flip attack, effectively mitigating the catastrophic threat compared to the FedAvg baseline. While this narrowly edges out Krum and FoolsGold estimates, the

Image Placeholder Advanced Attacks Bar Chart

Figure 10. Final test accuracy across advanced threat vectors for AEGIS.

Table II
FINAL ACCURACY (%) ACROSS BASELINES AND ATTACKS ($f = 30\%$, NON-IID)

Defense	No Attack	Sign-Flip	Label-Flip	ALIE
FedAvg	76.16	~ 10.0	~ 10.0	~ 10.0
Krum	71.30	~ 35.0	~ 35.0	~ 10.0
CWMed	72.31	~ 28.0	~ 28.0	~ 10.0
Bulyan*	67.69	N/A	N/A	N/A
FoolsGold	73.80	~ 65.0	~ 65.0	~ 10.0
AEGIS	76.08	67.38	75.13	17.39

*Bulyan mathematically requires $n \geq 4f + 3$. At $f = 30\%$, it violates this constraint and fails to execute under attack.

comprehensive benchmarking of these baseline defenses under optimization attacks remains ongoing to determine the definitive vulnerability gap.

E. Computational Complexity and Aggregation Runtimes

While AEGIS possesses an asymptotic complexity of $\mathcal{O}(kd)$, avoiding the $\mathcal{O}(k^2d)$ pairwise calculations of distance-based methods, our empirical profiling of the isolated server-side aggregation reveals significant constant factor overheads at small client scales.

By isolating the aggregation execution block via wall-clock timing across 500 rounds, the mean per-round aggregation times for $m \approx 20$ active clients were recorded as follows: FedAvg (~ 42 ms), Krum (~ 25 ms), CWMed (~ 30 ms), and AEGIS (~ 97 ms).

At this limited client scale, calculating a 20×20 pairwise distance matrix for Krum is highly vectorized and computationally trivial for modern GPUs. Conversely, AEGIS mandates multi-pass Introspect median operations across 290,000 parameter dimensions, followed by dense Median Absolute Deviation tensor calculations. These 1D operations carry a constant factor that dominates aggregation time at small scales, resulting in AEGIS requiring roughly four times the aggregation compute of Krum. AEGIS’s $\mathcal{O}(kd)$ advantage over $\mathcal{O}(k^2d)$ methods mathematically materializes only at massively distributed scales (e.g., cross-device deployments where $k > 100$), rather than small cross-silo configurations.

F. Resilience to Varying Byzantine Fractions

To evaluate the breaking point of the dual-metric filtering mechanism, we subjected the protocol to a standard Sign-Flip attack while scaling the global Byzantine fraction $f \in \{10\%, 20\%, 30\%, 40\%\}$.

Table III
AEGIS PERFORMANCE UNDER SIGN-FLIP ATTACK AT VARYING FRACTIONS

Byzantine Fraction (f)	Final Accuracy	Best Loss
0% (Baseline)	75.12%	0.7358
10%	73.16%	0.7811
20%	71.31%	0.8585
30%	66.79%	1.0347
40%	30.33%	2.2319

As illustrated in Table III, Aegis exhibits graceful degradation up to a 30% threat level. However, at $f = 40\%$, the protocol suffers a catastrophic collapse down to 30.33% accuracy. In a highly Non-IID environment (4 shards per client), a 40% coordinated adversarial presence, combined with the natural statistical skew of the remaining honest clients, is sufficient to overpower the two-pass median computation (Step 2). Once this median anchor is compromised, the Euclidean and Cosine gatekeepers are evaluated against a poisoned reference, causing systemic failure.

G. Sensitivity to Data Heterogeneity

We conducted a sensitivity analysis by varying the number of data shards allocated per client ($s \in \{2, 4, 6, 10\}$) under both benign and targeted attack conditions (Label-Flipping, $f = 30\%$).

Table IV
AEGIS ACCURACY ACROSS VARYING NON-IID DEGREES ($f = 30\%$)

Shards per Client	No Attack	Label-Flip	Accuracy Gap
2 (Extreme Skew)	68%	55%	13%
4 (Moderate Skew)	75%	71%	4%
6 (Mild Skew)	77%	74%	3%
10 (Near IID)	78%	76%	2%

As detailed in Table IV, extreme statistical heterogeneity ($s = 2$) creates natural variance that overlaps with malicious update distributions. This forces the aggregator into a tradeoff space where distinguishing malicious intent from legitimate rare-class updates becomes difficult, resulting in a lower baseline accuracy (68%) and a noticeable vulnerability gap (13%). At moderate to mild skew ($s \geq 4$), AEGIS successfully separates adversarial vectors from natural variance via its two-pass system, minimizing the accuracy degradation to a marginal 2-4%.

VII. LIMITATIONS

While AEGIS mitigates standard attacks, our evaluation confirms fundamental limits consistent with theoretical impossibility results [5]:

- 1) **Variance-Envelope Attacks:** Omniscient ALIE bypasses per-round statistical filtering entirely. Attackers pass both Cosine and Euclidean checks (0.0% empirical detection rate), bringing the model to 17.39% accuracy.
- 2) **Small- ϵ IPM:** While two-pass decontamination catches large- ϵ IPM, small- ϵ manipulations evade detection.
- 3) **Reputation Bootstrap Failure:** Under attacks like ALIE, the global model diverges before the EMA reputation signal ($\gamma = 0.95$) accumulates sufficiently to isolate the compromised nodes.
- 4) **Sybil Thresholds:** Coordinated Sybil attacks inflating the Byzantine fraction to 40% or beyond overwhelm the median-based anchors, dropping utility catastrophically (e.g., 30.33% accuracy at $f = 40\%$).

VIII. CONCLUSION AND FUTURE WORK

Aegis provides a robust protocol for securing FL against conventional poisoning and volume spam in highly Non-IID domains. It utilizes a two-pass median decontamination and cross-round reputation tracking system to mitigate directional and magnitude anomalies. Future work must bridge the gap exposed by ALIE. We plan to explore integration with spectral methods to detect correlated perturbations, and evaluate AEGIS at larger client scales ($k > 100$) where its $\mathcal{O}(kd)$ asymptotic complexity can overcome the computational overhead of multi-pass tensor operations.

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